vector space (linear space).

subspace sum, intersection

linear (in)dependent

basis, standard basis (e.g. infinite-dim).

coordinate existence and uniqueness.

$\Rightarrow$ any finite-dim vector space is isomorphic to $\mathbb{R}^n$.

dimension (independent from the choice of basis).

direct sum

infinite-dimensional vector spaces

$0 \leq n \leq \infty$.

$\mathbb{R}[x]$, $\mathcal{F}(\mathbb{R}, \mathbb{R})$, $\mathbb{R}^{\mathbb{N}}$, $\mathbb{R}$ (over $\mathbb{Q}$), $C^n(\mathbb{R})$.

$\{x^n\}$
$0 \leq n \leq \infty$.

$\{e_1, e_2, e_3, \cdots\}$ is not a basis, finite combination.

direct sum.

$\mathbb{R}^2 = \langle e_1 \rangle \oplus \langle e_2 \rangle$, $\mathbb{R}[x] = \{f(x) : f(-x) = f(x)\} \oplus \{f(x) : f(-x) = -f(x)\}$

$M_n(\mathbb{R}) = \{A : A^T = A\} \oplus \{A : A^T = -A\}$.

$\dim = \frac{n(n+1)}{2}$ $\dim = \frac{n(n-1)}{2}$

$$\mathbb{R}[x] = \bigoplus_{n=0}^{\infty} \langle x^n \rangle.$$

note $\bigoplus_{n=1}^{\infty} \mathbb{R} \subsetneq \prod_{n=1}^{\infty} \mathbb{R} = \mathbb{R}^{\infty}$.